

Cheap Geometric Priors for Viewpoint- and Pose-Robust VLAs

An Injection Architecture, a Robustness Fingerprint, and a Measured Collapse

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Abstract

Vision-Language-Action (VLA) models achieve near-perfect success on clean manipulation benchmarks but degrade sharply when the camera moves or the robot starts from an unfamiliar pose. This project asks a cost-focused question the literature leaves open: **how much robustness can the *cheapest possible* geometric prior buy?** I injected a single geometry token — derived from free simulator depth, converted to a world-frame point cloud and encoded by a ~18M-parameter PointNet — into OpenVLA-OFT (7B parameters), and measured per-perturbation-axis robustness on the LIBERO-Plus benchmark (2,519 perturbed tasks).

Three concrete outcomes. **(1) A per-axis fragility fingerprint** of the base model showing selective collapse on geometric perturbation axes (Camera Viewpoints 45.8%, Robot Initial States 34.6%) while appearance axes hold (Background 82.0%, Language 96.3%). **(2) A verified end-to-end injection architecture:** a step-10 checkpoint — geometry token zero-initialized, action head frozen, LoRA effectively null — passed through the *entire* custom pipeline (data, surgery, merge, save, eval) and scored **80% closed-loop**, proving every subsystem correct and the token harmless at initialization. **(3) The central measured finding:** any *effective* fine-tuning of this model on its 500 source demonstrations destroys closed-loop competence within roughly **250 gradient steps (~4,000 samples, <3% of one epoch)** — success collapsing from 80% to 0% — while offline action-prediction error *improves monotonically* over the same interval. The result was established by systematic elimination across six candidate explanations, including a false lead (checkpoint file corruption) that was itself ruled out by checksum forensics, a single-file transplant experiment, and a controlled solo replication. The offline/closed-loop divergence is a concrete, self-generated demonstration of compounding covariate shift — arguably the central evaluation trap in imitation learning — measured here with unusual evidentiary rigor.

1. Motivation and Fragility Fingerprint

VLAs inherit their perception from 2D vision-language pretraining, and prior work (PointVLA, arXiv:2503.07511; PointACT, arXiv:2605.21414; Any3D-VLA, arXiv:2602.00807) shows that adding 3D information improves manipulation — but those methods sit at the expensive end of a design spectrum: pretrained point transformers, multi-layer injection blocks, action-expert fusion. Nobody has mapped the *cost-vs-robustness frontier*: what does each additional dollar of 3D machinery buy, per perturbation axis? This project built the cheapest rung of that ladder and set out to measure it.

Gate 1 — reproducing the baseline. I first reproduced OpenVLA-OFT's published performance on clean LIBERO-Spatial: **490/500 = 98.0%** over 500 trials on WashU's SLURM cluster (A40 GPUs).

Gate 1b — the fingerprint. I then evaluated the same checkpoint on the LIBERO-Plus perturbed libero_10 suite (2,519 task variants across seven perturbation categories), producing the motivating measurement:

Perturbation Axis	Base Model Success	Tasks
Camera Viewpoints	45.8%	419
Robot Initial States	34.6%	393
Background Textures	82.0%	289
Language Instructions	96.3%	383
Sensor Noise	61.1%	185 (partial)

Base OpenVLA-OFT on LIBERO-Plus: selective geometric fragility

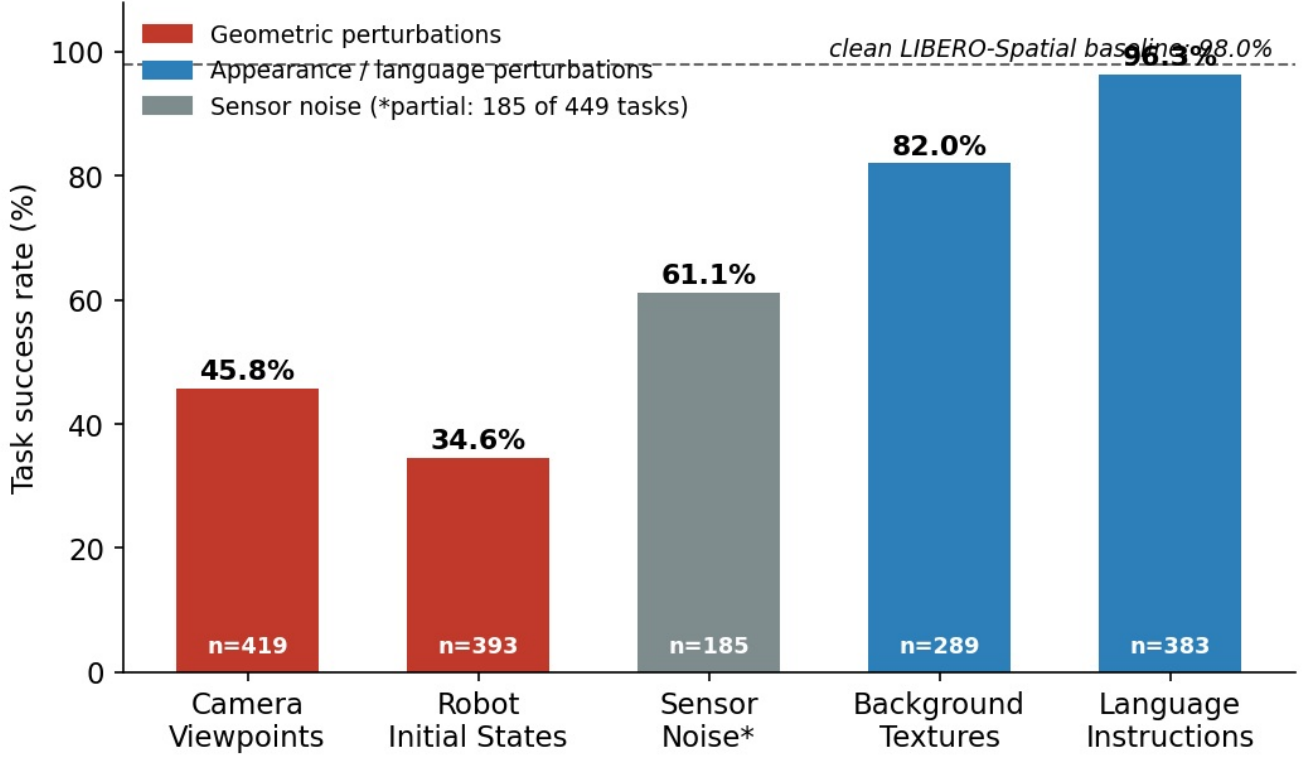


Figure 1. Base OpenVLA-OFT per-axis success on the LIBERO-Plus perturbed suite. Geometric perturbations (red) collapse performance while appearance and language perturbations (blue) largely preserve it; the dashed line marks the 98.0% clean-benchmark baseline.

The collapse is *selective and geometric*: the model tolerates appearance and language changes but loses half or more of its competence when the viewpoint or robot pose shifts. This defined the project's hypothesis: **a world-frame point cloud token should preferentially recover the camera and robot-init axes**. The fingerprint stands as an independent, citable result regardless of what followed.

2. System Architecture

2.1 Data pipeline: free depth → world-frame point clouds

Because LIBERO is a simulator, depth is free. A replay pipeline re-executes all 500 demonstration trajectories through MuJoCo's offscreen renderer, capturing per-frame RGB, depth, camera intrinsics **K**, and extrinsics **R** (`replay_with_depth.py`). A second stage (`make_pointclouds.py`) back-projects depth to 3D, transforms to the world frame, crops to an empirically derived workspace box ($x \in [-0.75, 0.35]$, $y \in [-0.70, 0.70]$, $z \in [0.82, 1.42]$, via `derive_workspace_box.py`), and randomly samples **512 points per frame**. Output: ~134,590 (frame, point-cloud) pairs as compressed NumPy archives. The eval-time live rendering path was validated against this pipeline to **0.015 m mean per-axis agreement** with zero out-of-workspace padding.

Geometry-token injection pipeline: free sim depth → one token in the VLA sequence

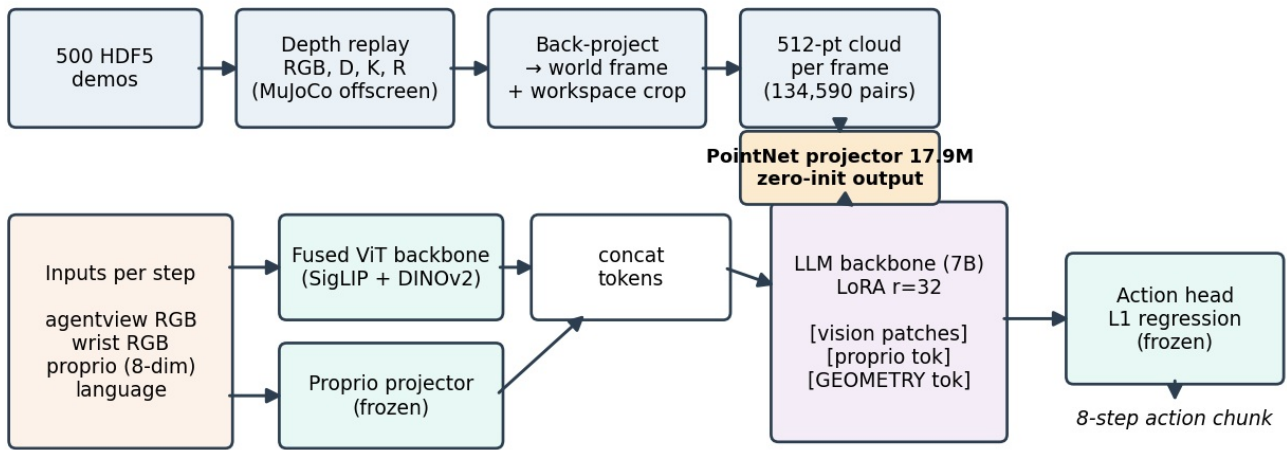


Figure 2. The injection pipeline. Top: demonstration replay captures depth, intrinsics, and extrinsics; back-projection and workspace cropping yield 512-point world-frame clouds for all 134,590 frames. Bottom: the zero-initialized PointNet projector contributes one geometry token to the LLM sequence beside the vision patches and proprio token; the action head and proprio projector remain frozen.

2.2 Model surgery: one token into a 7B VLA

OpenVLA-OFT's HuggingFace implementation (`modeling_prismatic.py`) was modified to accept a point cloud in both training and inference paths:

- **PointCloudProjector** (~17.9M parameters): a PointNet-style encoder mapping (512, 3) → one 4096-dim token.
- **`_process_geometry_features`**: appends the geometry token to the projected vision patch embeddings beside the proprio token in both `forward` and `predict_action`, with `NUM_PATCHES` bookkeeping updated so action hidden-states are extracted from the correct positions.
- **Zero-initialized output layer** (PointVLA-style): the projector contributes an exact zero vector at step 0, so the fine-tune's first forward pass is behaviorally identical to the pretrained base model. This design choice was later *verified behaviorally*: the step-10 checkpoint scored 80% closed-loop (§4.1).

2.3 Custom training dataset

`LiberoGeomDataset` + `GeomPaddedCollator` join, per timestep: agentview RGB, wrist RGB, an 8-dim normalized proprio vector (EE position + axis-angle orientation + gripper state, bounds-q99 normalized verbatim to the eval path's `normalize_proprio`), the point-cloud frame, the language instruction, and an 8-step action chunk with OFT's gripper convention (`gripper = 1 - clip(gripper, 0, 1)`) replicated from the RLDS `libero_dataset_transform`. A `--shuffle_points` flag supports a random-geometry control rung; `--use_geom_dataset` decouples the dataset from token injection so the RGB-only baseline trains on byte-identical data.

2.4 Training recipe (frozen-component)

Converged on after two failed recipes (§4):

Component	Setting	Rationale
Trainable	LoRA r=32 (~111M) + PointCloudProjector (~18M)	adapt, don't retrain
Frozen	pretrained action head (151M) + proprio projector	preserve gripper timing (PointVLA / two-phase adaptation recipe) — freeze verified bit-exact across saved checkpoints
LR	5e-5	OFT's refinement-phase rate; 5e-4 churns converged components
Batch	4 × grad-accum 2 (effective 8)	fits 2-image input on a 44 GB A40
Init	pretrained action head + proprio projector loaded; projector zero-init	start from the working policy

Treatment (geometry) and control (RGB-only) rungs were trained with identical recipes, differing only in the token.

3. Verification Methodology

The project's most transferable output is a set of diagnostic instruments for VLA train/eval pipelines, each of which proved decisive at least once:

1. **Base-model-through-same-eval control.** Running the known-good pretrained checkpoint through the identical eval invocation separates "my model is broken" from "my eval is broken."
2. **Near-null-training identity check.** A checkpoint saved at step 10 — where zero-init + frozen components + near-zero LoRA make the model behaviorally the base model — tests the *entire* pipeline (data loading, surgery, LoRA merge, checkpoint save, eval load) in one experiment. Its 80% result was the project's single most information-dense measurement.
3. **Offline prediction probe.** A standalone script loads any checkpoint (model + heads + projectors, handling DDP prefixes and norm-stats injection), feeds frames *from its own training data*, and compares predicted action chunks to ground truth — bypassing the simulator. Its per-checkpoint gripper-anticipation readout became the convergence criterion the loss curve could not provide.
4. **Input capture and pixel forensics.** Instrumenting the eval loop to dump the exact observation tensors the model receives, then comparing them numerically (per-axis correlations under candidate transforms) and visually against training frames.
5. **Component checksum forensics and transplant testing.** Exploiting the frozen-component invariant (frozen weights must be numerically identical across all runs) to fingerprint anomalies; then swapping a single suspect file between checkpoints to convict or exonerate it functionally. A one-file transplant flipped a 0% checkpoint to 70%.
6. **Controlled solo replication.** Re-running a configuration in isolation (no concurrent jobs, verified data on disk) as the final arbiter between artifact and phenomenon.
7. **Per-axis result slicing** (`slice_by_category.py`), converting raw episode logs into per-perturbation-category tables.

Instrument-first debugging: identical symptoms, orthogonal instruments

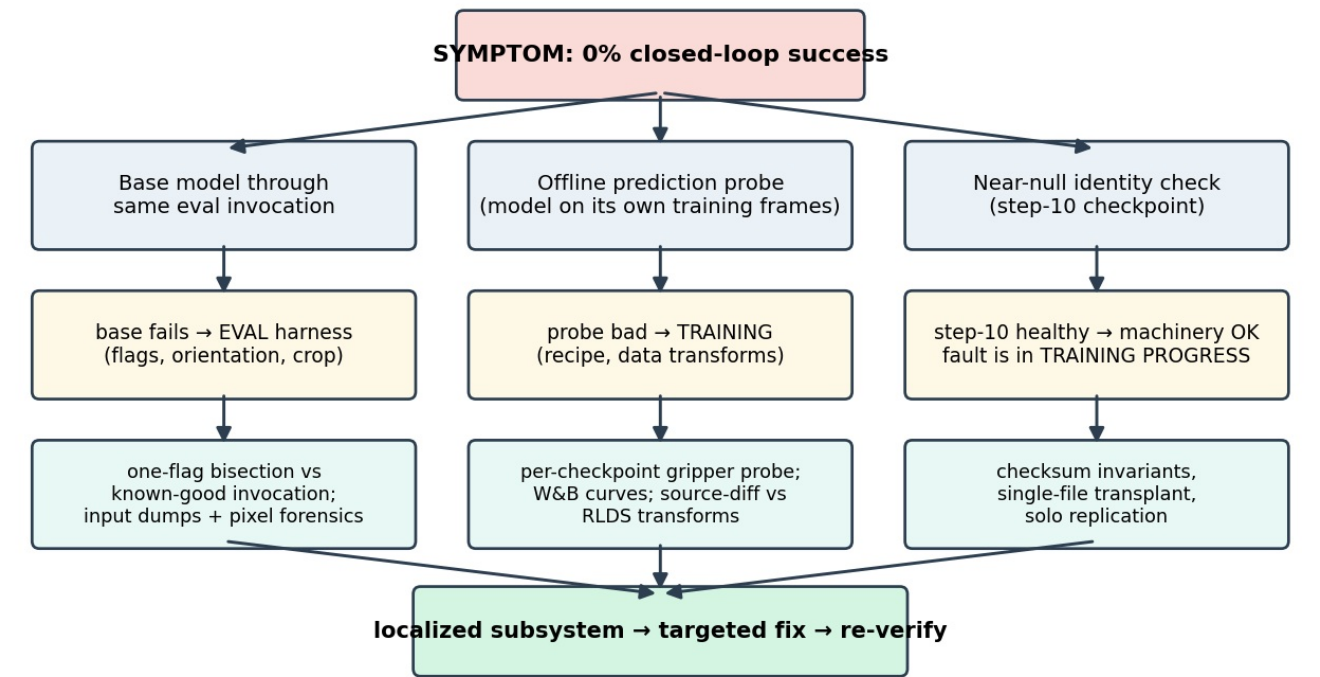


Figure 3. The instrument-first debugging methodology. A single undifferentiated symptom (0% success) is factored by three orthogonal instruments into eval-harness, training, and training-progress fault classes, each with its own targeted forensic toolset.

4. Findings

4.1 The pipeline works: the step-10 identity result

A checkpoint trained for only 10 gradient steps — passing through every custom component — scored **80% closed-loop** on the leading (camera-perturbation) block of the perturbed suite, matching base-model-level performance. This single experiment simultaneously validated: the custom dataset, the model surgery, the geometry token's zero-init harmlessness, the LoRA merge, the checkpoint save/load path, and the evaluation harness. Every subsequent result rests on this foundation: whatever failed later, it was not the machinery.

4.2 The failure-mode ledger

Every failure produced the same symptom — 0% task success — making the failures indistinguishable without instruments. In the order discovered:

#	Issue	Symptom	Instrument	Resolution
1	Gripper convention: raw HDF5 actions (-1 =open/ $+1$ =close) vs OFT's binarized (1=open/0=close)	inverted grasping, 0%	reading the RLDS transform source against the custom dataset	replicate 1 – <code>clip(g,0,1)</code>
2	Eval-flag/model-config mismatch: <code>--num_images_in_input 1</code> passed to the natively 2-image base model	0% for every model incl. base	base-model control + one-flag bisection against a known-good invocation	eval flags must match each model's training config
3	Optimization recipe: LR 5e-4 on warm-started converged components	flat noisy L1 ≈ 0.2 ; gripper degraded from timing → mean-hedging → state-copying	W&B curves + offline probe across checkpoints	LR 5e-5 + frozen head/projector + zero-init
4	Image orientation: eval 180°-flips renders (matching the RLDS-regenerated data the base model used); custom training used raw HDF5 orientation	garbage motion (robot dives into floor)	input-dump pixel forensics; confirmed by a source-code comment	per-model <code>flip_images</code> eval flag
5	Silent patch failure: a dataset patch script died on an anchor assertion 9 seconds before job submission; two multi-day "flipped-data" training campaigns actually trained on unflipped data	experiments testing a hypothesis that was never actually run	filesystem mtime forensics vs SLURM job-start timestamps	gated launches: verify patches on disk (grep-confirmed) before any submission
6	A false lead, fully prosecuted: two nominally identical 250-step runs produced an 84% checkpoint and a 0% checkpoint; checksum forensics isolated the difference to a single file (the pointcloud projector), and transplanting the healthy run's file flipped 0% → 70% — initially suggesting checkpoint corruption under concurrent training. Numerical autopsy then showed <i>both</i> files statistically healthy (no NaN/Inf, matched norms), and training was shown to be non-deterministic across runs. Revised interpretation: seed-level variance near the collapse threshold — at ~250 steps, one run's projector had crossed into the harmful regime and the other had not.	contradictory 84%-vs-0% at identical step counts	md5 sweeps over frozen-component invariants; single-file transplant; weight-statistics autopsy	resolved by the solo replication (§4.3)

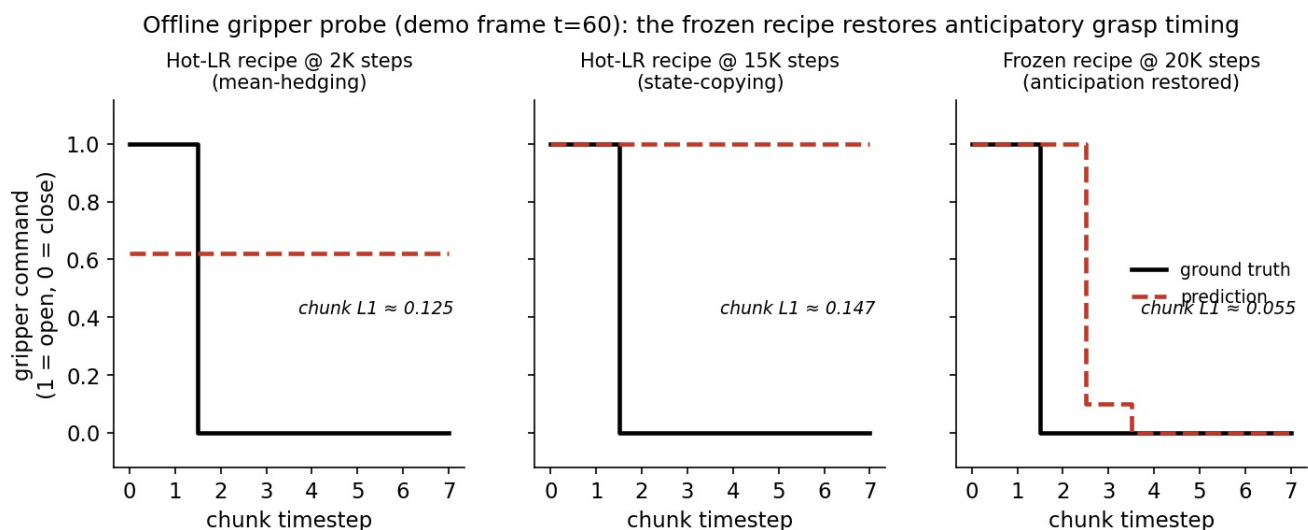


Figure 4. Offline gripper probe on a held training frame (demo_0, $t=60$) across recipes. The hot-LR recipe degrades from mean-hedging (2K steps) to state-copying (15K); the frozen recipe recovers anticipatory closure one timestep after ground truth — the hardest behavioral signal in the data — while cutting chunk L1 error 2-3 $\times$.

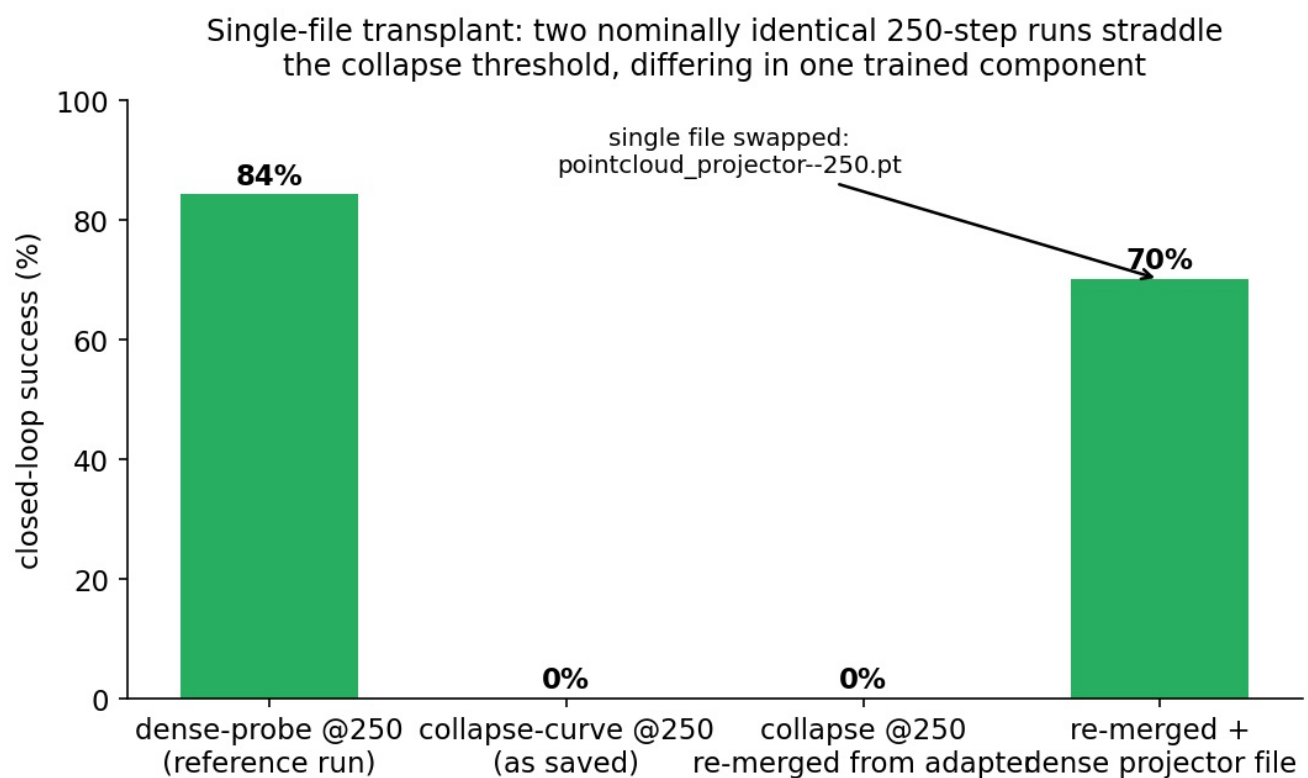


Figure 6. The single-file transplant experiment (failure-mode ledger, row 6). Two nominally identical 250-step runs produced an 84% and a 0% checkpoint; re-merging the failing checkpoint cleanly from its adapter changed nothing, but swapping in the healthy run's pointcloud-projector file recovered 70% — isolating the behavioral difference to one trained component and, after a clean weight-statistics autopsy of both files, establishing seed-level variance near the collapse threshold rather than file corruption.

4.3 The central finding: effective fine-tuning collapses closed-loop competence within ~250 steps

The final controlled experiment — a **solo** training run on **verified-flipped** data (matching the base model's visual convention, so learning is maximally effective), with the full frozen recipe — collapsed to **0% by its first saved checkpoint (step 250)**, evaluated through the standard harness that scores the base model at 46-96% by axis.

The complete evidence chain:

- **Step 10 (near-null training): 80%.** The machinery is sound; the token is harmless.

- **Unflipped 250-step runs: 75-92%** (dense-probe series). Healthy — but these models trained *against* a 180° visual-convention mismatch with the base model, so their effective learning was minimal; they remained near base behavior.
- **Unflipped runs at 1,000-20,000 steps: 0%** across two recipes and both rungs (with and without the geometry token — the paired RGB control collapsed identically, exonerating the token).
- **Flipped (convention-matched) solo run: 0% by step 250.** When learning is effective, collapse is nearly immediate.
- **Throughout, offline prediction improved:** the 20K frozen-recipe checkpoint predicted its own training frames at L1 0.036-0.074 with correct anticipatory gripper transitions — the hardest signal in the data — while scoring 0% in closed loop, *including on unperturbed-scene language-instruction variants* where the base model scores 96.3%.

Interpretation. This is compounding covariate shift in its sharpest form. Five hundred demonstrations cover a thin tube of state space; the base model's robustness came from regenerated, augmented data at 100× the optimization budget. Fine-tuning — even parameter-efficient, even with the action decoder frozen — shifts the LLM's hidden-state distribution away from what the frozen head was calibrated for and narrows the policy toward demo memorization. In closed loop, the policy's own small errors carry it off the demo manifold within a few contact-rich steps, and errors compound where the base model recovers. Rollout video shows the signature: correct task sequencing (approach the stove first), degrading precisely at the first fine-contact moment. The measured statement: **closed-loop success fell from 80% to 0% within ~250 effective gradient steps (~4,000 samples, <3% of one epoch) while offline action error improved monotonically.**

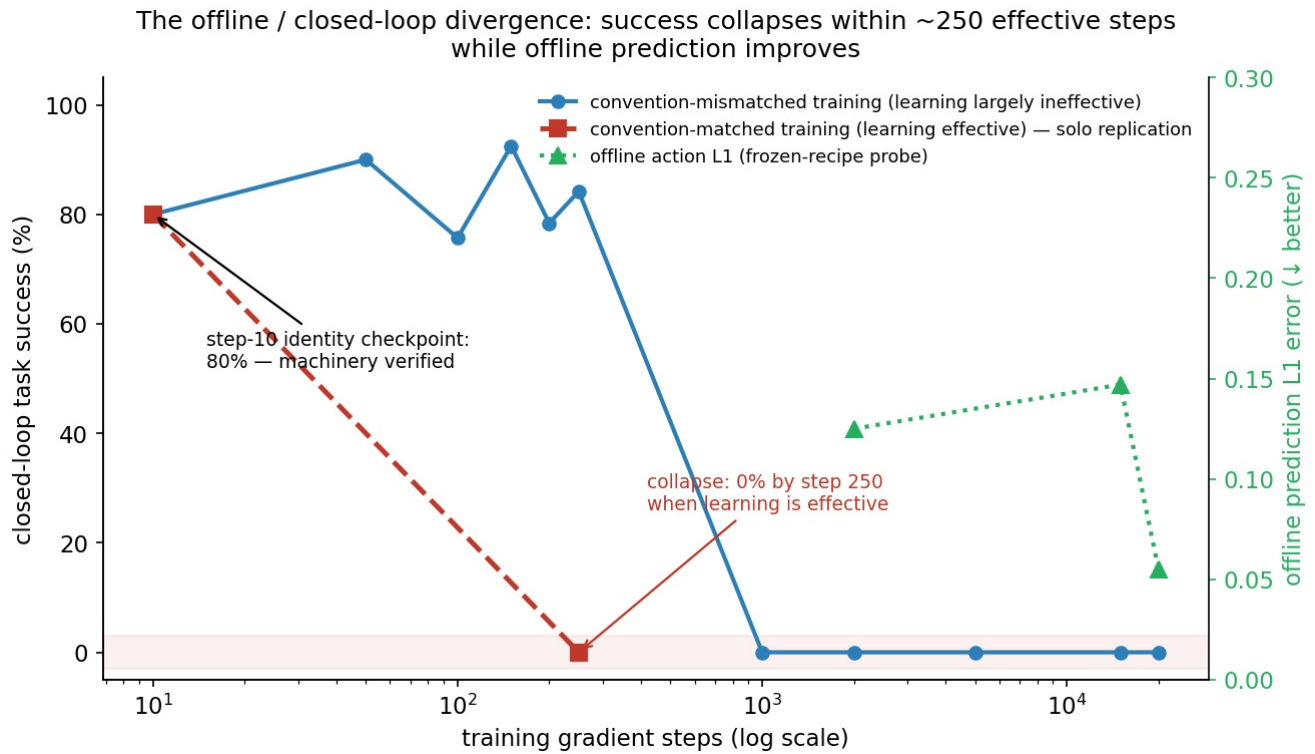


Figure 5. The central result. Closed-loop success (left axis) against training steps, log scale. Convention-mismatched runs (blue), whose learning was largely ineffective, remain near the 80% step-10 anchor through 250 steps and collapse only with longer training; the convention-matched solo replication (red) collapses to 0% by its first checkpoint at step 250. Offline prediction error (green, right axis) improves over the same interval — the divergence of the two measures is the finding.

4.4 Secondary findings

- **Zero-init injection is behaviorally safe** (80% at initialization) — a verified claim about the architecture, consistent with PointVLA's design argument.
- **The frozen-component recipe restores offline gripper competence** that hotter recipes destroy (anticipation recovered; offline error cut 2-3×) — necessary but, per §4.3, not sufficient for closed-loop transfer.
- **Training-outcome variance near the collapse threshold:** nominally identical 250-step runs can straddle the healthy/collapsed boundary (84% vs 0%), differing in a single trained component. Single-seed results in this regime are not trustworthy.
- **Offline metrics, including per-dimension behavioral probes, do not certify closed-loop competence** — the probe showed textbook-converged behavior on checkpoints that scored zero.

5. Results Summary and the Open Frontier

The cost-robustness frontier question — does a cheap geometry token preferentially recover geometric perturbation axes? — remains open, because no fine-tuned model (with or without the token) survived closed-loop evaluation long enough to measure. What is settled: the injection architecture is sound and safe; the fragility fingerprint motivating the question is real; and the barrier is now precisely characterized — **narrow-data fine-tuning collapse** — with a specified fix: train from the RLDS-regenerated, augmentation-matched dataset the base model used (eliminating the entire preprocessing-mismatch class by construction) at base-recipe scale (50K+ steps, large effective batch), and/or interleave base-distribution data as rehearsal, and/or adopt representation-level geometry injection (Spatial-Forcing-style alignment) that never perturbs the action pathway. Each is a multi-GPU-week undertaking deliberately out of scope for this project.

6. Technical Skills and Tooling

Large-model engineering. Surgical modification of a 7B-parameter HuggingFace `trust_remote_code` model (token injection in training and autoregressive inference paths; patch-count bookkeeping; config/auto-class management across HF snapshot caches); LoRA fine-tuning and adapter merging (PEFT `merge_and_unload`, offline re-merge from adapters); selective freezing with numerical verification; warm-start loading of DDP-checkpointed components (state-dict prefix handling, bf16 conversion); PointNet encoder implementation with zero-initialization.

Robot learning. Behavior cloning with action-chunked continuous control (L1 regression heads); action-space conventions and normalization (bounds-q99, gripper binarization); proprioception construction and verification (quaternion → axis-angle equivalence checks); the OpenVLA-OFT recipe (LR schedules, convergence criteria, multi-camera fused backbones); benchmark evaluation on LIBERO / LIBERO-Plus; the offline-vs-closed-loop distinction and covariate shift, measured firsthand.

3D geometry. Depth back-projection with camera intrinsics/extrinsics; world-frame transforms; workspace derivation and cropping; point-cloud sampling; cross-path render validation (MuJoCo/robosuite offscreen rendering, EGL, orientation conventions).

HPC & experiment operations. SLURM job design (sbatch, dependencies for sequential pipelines, resource/walltime budgeting, `sacct/squeue` forensics); conda environment management across incompatible training/eval stacks; CUDA OOM diagnosis and mitigation (gradient accumulation, allocator configuration); multi-day training campaigns; Weights & Biases tracking; storage management across home/scratch/project filesystems and HuggingFace caches; filesystem-timestamp forensics for reconstructing experiment provenance.

Debugging methodology. Controlled one-variable bisection against known-good references; instrument-first debugging; paired treatment/control design; near-null identity checks as whole-pipeline tests; checksum invariants and single-file transplant experiments; pixel- and tensor-level input forensics; programmatic code patching with anchored, idempotent, syntax-verified edit scripts — and gated launches after a silent patch failure cost four GPU-days.

Software & tools: PyTorch (DDP, `torch.distributed.run`), HuggingFace Transformers, PEFT/LoRA, safetensors, OpenVLA-OFT, LIBERO / LIBERO-Plus, robosuite/MuJoCo, Weights & Biases, SLURM, conda, h5py, NumPy, PIL/OpenCV, `tf.data`/RLDS (read-side), draccus.

7. Lessons Learned

1. **Train/eval configuration consistency is the dominant failure class in VLA fine-tuning.** Four independent silent mismatches produced identical 0% symptoms, each in a different subsystem.
2. **Eval flags are per-model, not per-benchmark.** An invocation correct for one checkpoint silently breaks another trained with different inputs.
3. **Build the offline probe and the near-null identity check before the first real experiment.** Together they cost two hours and cleanly factor the system into model / machinery / training-dynamics.
4. **Always run the paired control.** The RGB-only rung exonerated the geometry token at every stage — a conclusion no single-arm experiment could reach.
5. **Offline metrics do not certify closed-loop behavior.** Not as a footnote: an 80-point closed-loop swing occurred against a monotonically improving offline metric.
6. **Verify patches on disk before launching jobs.** A patch script that died 9 seconds before submission silently invalidated two multi-day campaigns; every launch is now gated on `grep`-confirmed file state.
7. **In high-variance regimes, one seed is not a result.** Nominally identical runs straddled the healthy/collapsed boundary; replication is part of the measurement.
8. **Chase anomalies to mechanism, but pre-commit stopping rules.** The 84%-vs-0% contradiction was worth prosecuting; the pre-committed rule ended the project at a conclusion, not an exhaustion.